TONER-1 Pro

12,194,688 parameters
int8 weights, one scale per row
Inference runs inside the PostScript interpreter

State of the art among language models that execute inside a print job.

SAMPLE OUTPUT

Unedited model output. No prompt: generation starts at the BOS token.

Once upon a time, a little bear sat on a table near a printer. The printer had a red one. The bear wanted a new friend, but he did not have a picture of a cake. The printer was sad. The printer beeped softly and blinked. “I am four years old,” the printer said. “Do you want me to eat you?” the teddy bear nodded. The printer took a bite of honey. It made a funny face. The bear laughed. “That is my cake!” he said. Then the printer heard a loud noise. It was the printer. The printer was printing a big cake. The bear wanted to eat the cake. The printer stopped beeping. The printer beeped once. “I am sorry,” the printer said. The bear came to help. The bear threw another piece of cake on the floor. The printer ate another piece. Each page was good at printing. The printer hummed and printed the cake. Everyone was happy. The printer learned to…

TOKEN STREAM

The first 70 of 200 sampled tokens, one box per vocabulary piece. · marks a space, ¶ a newline.

·Once	·upon	·a	·time	,	·a	·little	·bear	·sat	·on	·a	·table	·near	·a
·printer	.	·The	·printer	·had	·a	·red	·one	.	·The	·bear	·wanted	·a	·new
·friend	,	·but	·he	·did	·not	·have	·a	·picture	·of	·a	·cake	.	·The
·printer	·was	·sad	.	·The	·printer	·beeped	·softly	·and	·blinked	.	·"	I	
·am	·four	·years	·old	,"	·the	·printer	·said	.	·"	Do	·you	·want	·me	·to

HOW THIS PAGE WAS MADE

1 HOST
Sends the print job. Performs no arithmetic.

→

2 INTERPRETER
Parses the int8 weights, then runs the forward pass once per token.

→

3 PAGE
The same job typesets the story and calls showpage.

TOKENS	SECONDS	TOKENS / SEC	SEED	TEMPERATURE
200	**71.1**	**2.81**	**42**	**0.8**

INTERPRETER	GPL Ghostscript · revision 10071 · PostScript version 3010
MODEL	TONER-1 Pro 12M (Llama-2 architecture) · dim 384 · 6 layers · 6 query / 6 KV heads · 4,096-token vocab
STOPPED BY	Token budget: STEPS = 200. The story was cut off mid-thought.

Host compute: 0 FLOPs. Inference cost includes toner.

Every word above was generated on-device, one token at a time, by the interpreter named in this footer; the host computer only sent bytes. Seconds are interpreter usertime spent generating. rand is implementation-defined, so the same seed on a different interpreter tells a different story.